AI HISTORY TIMELINE · 1943 TO TODAY

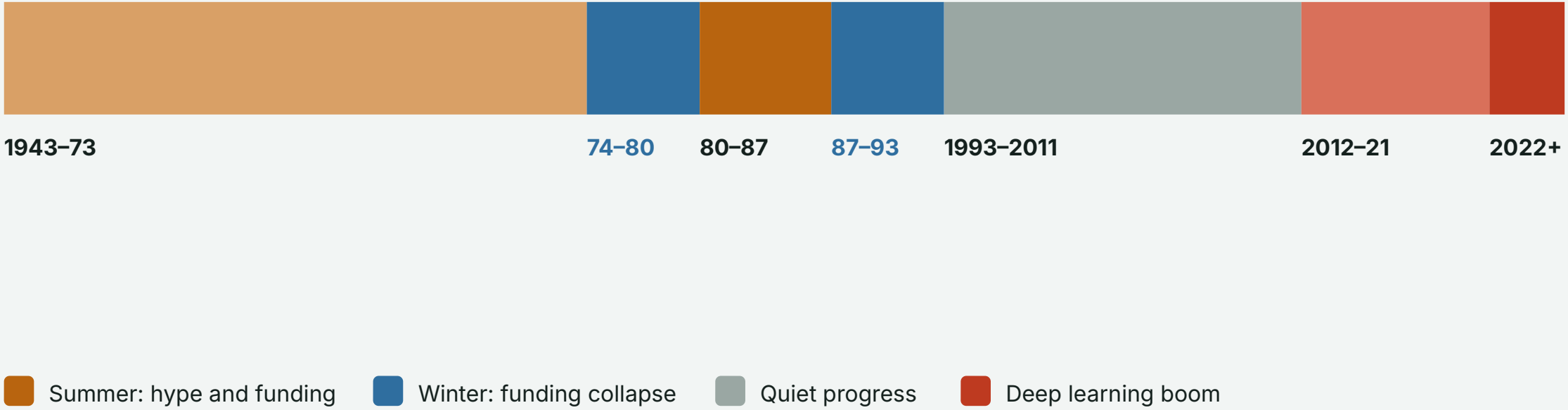
Seasons of AI

Eighty years of summers, winters and breakthroughs

Akash Singh · AIML-500: AI Lab · October 2026

AI history is a weather report

Progress came in seasons. Summers brought hype and funding. Winters arrived when promises outran results.



1943 – 1973 · SUMMER

The birth of AI

1950

Turing Test

Alan Turing asks “Can machines think?” and proposes the imitation game.

1956

Dartmouth Workshop

John McCarthy, Marvin Minsky and Claude Shannon coin “artificial intelligence.”

1958

Perceptron

Frank Rosenblatt builds the first neural network that learns from examples.

1966

ELIZA

Joseph Weizenbaum’s chatbot fools users with simple pattern matching.

The warning sign, 1969: Minsky and Papert’s book *Perceptrons* shows single-layer networks can’t learn simple functions like XOR. Neural-network funding dries up.

1974 – 1980 · WINTER

The first AI winter

Causes

- Overpromising: human-level AI “within 20 years”
- Weak hardware with kilobytes of memory
- Toy solutions failed on real-world problems
- 1973 Lighthill Report: UK cuts funding, DARPA follows

Impact

- Grants vanish and labs shrink
- Neural-network research nearly stops
- Researchers avoid the label “AI” on their work

1980 – 1987 · SUMMER

The expert systems boom

1980

XCON at DEC

A rule-based expert system configures computer orders and saves Digital Equipment Corp. millions a year. AI becomes an industry.

1982

Japan’s Fifth Generation

A national push for intelligent computers sets off a funding race in the US and UK.

1986

Backpropagation

Rumelhart, Hinton and Williams show how to train multi-layer neural networks. Deep learning still uses this method.

1987 – 1993 · WINTER

The second AI winter

Causes

- Cheaper Apple and IBM desktops kill the Lisp machine market
- Expert systems prove brittle and costly to update
- Japan’s Fifth Generation misses its goals
- DARPA cuts AI funding again

Impact

- Hundreds of AI companies close
- “AI” becomes a word to avoid in grant proposals
- Progress continues quietly: in 1989 Yann LeCun’s CNN reads handwritten ZIP codes

1993 – 2011 · QUIET PROGRESS

Three ingredients come together

Algorithms

1997 · LSTM
Hochreiter and Schmidhuber give networks a memory for sequences.

2006 · Deep belief nets
Geoffrey Hinton revives the term “deep learning.”

Data

2009 · ImageNet
Fei-Fei Li’s dataset grows to over 14 million labeled images.

The internet
Billions of web pages and photos make training data abundant.

Compute

2006 · Amazon Web Services
Cloud computing power can be rented by the hour.

2007 · NVIDIA CUDA
Gaming GPUs become engines for parallel math.

Headlines of the era: IBM’s Deep Blue beats Kasparov (1997). Watson wins *Jeopardy!* and Apple launches Siri (2011).

2012 · THE TURNING POINT

Hardware finally caught up with theory

≈15%

AlexNet’s top-5 error

26%

Next-best entry

Alex Krizhevsky, Ilya Sutskever and Geoffrey Hinton trained AlexNet on two NVIDIA gaming GPUs and won the ImageNet challenge by a landslide.

2014 – 2021 · DEEP LEARNING REVOLUTION

Deep learning takes over

2014

GANs

Ian Goodfellow's two competing networks generate realistic images.

2015

OpenAI

Founded as a nonprofit lab with $1 billion pledged.

2016

AlphaGo

DeepMind beats Go champion Lee Sedol 4–1.

2017

Transformer

Google's "Attention Is All You Need" becomes the "T" in GPT.

2020

GPT-3

175 billion parameters. AlphaFold 2 predicts protein shapes.

2018: Hinton, LeCun and Bengio win the Turing Award as the "godfathers of deep learning."

2022 – TODAY · GENERATIVE AI

Generative AI goes public

Year	Milestone	Why it matters
2022	ChatGPT launches on November 30	About 100 million users in two months
2023	GPT-4, Anthropic’s Claude, Meta’s Llama, Google’s Gemini	The multimodal model race begins
2024	Nobel Prizes for Hinton, Hopfield, Hassabis and Jumper	AI recognized as science; EU AI Act passes
2025	DeepSeek-R1, reasoning models and AI agents	Cheaper frontier AI; NVIDIA hits $5 trillion

Two ideas that frame the story

SYMBOLIC → CONNECTIONIST

Rules vs. learning

Early AI hand-coded logic and rules, from expert systems to Deep Blue. Modern AI learns patterns from data with neural networks.

NARROW AI VS. AGI

Specialist vs. generalist

Every system in this timeline is narrow AI. Artificial General Intelligence, matching humans at any task, is still a goal and a debate.

TODAY

The leading players

Models

- **OpenAI:** GPT and ChatGPT
- **Google DeepMind:** Gemini, AlphaFold
- **Anthropic:** Claude
- **Meta:** Llama, open weights
- **DeepSeek:** open reasoning models

Chips

- **NVIDIA:** GPUs and CUDA
- **AMD:** GPUs
- **Google:** TPUs

Cloud

- Microsoft Azure
- Amazon Web Services
- Google Cloud

Training compute for top models has doubled roughly every six months since 2010 (Our World in Data).

Three lessons from 80 years

Hype has a cost

Both winters followed promises AI couldn’t keep. Today’s boom should remember that.

Ideas wait for hardware

Neural networks (1958) and backpropagation (1986) only won once GPUs, cloud and big data arrived.

People drive progress

From Turing to Hinton to Fei-Fei Li, persistent researchers carried the field through the winters.

Sources

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AI use disclosure: I used Claude (Anthropic) to help research milestones and design these slides. I checked the events against the sources above.